

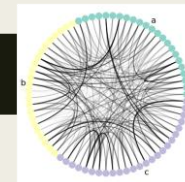
NETWORK ANALYSIS



matplotlib



NetworkX
Network Analysis in Python

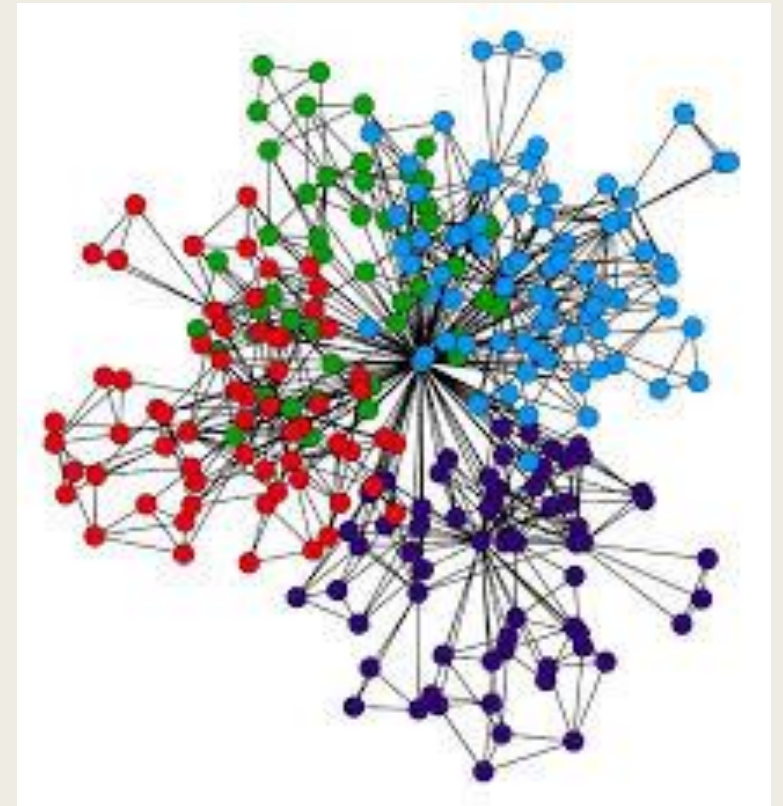


Outline

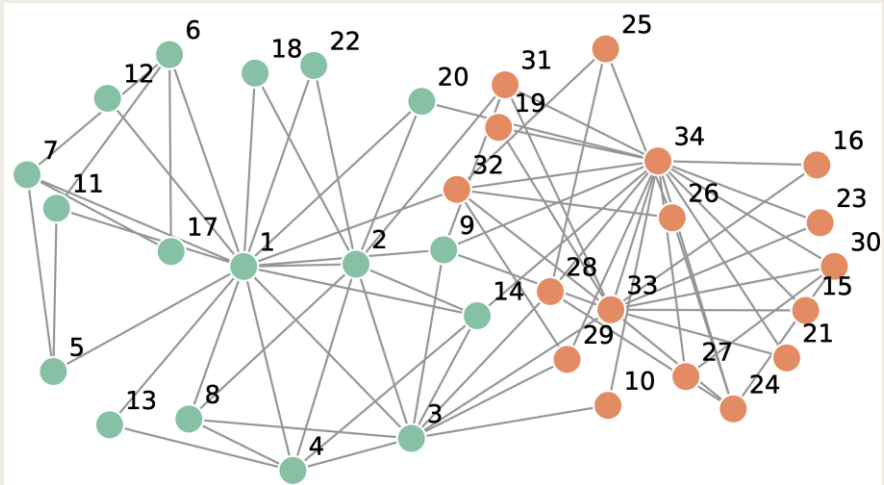
- Networks around us
- Applications
- Types of networks
- Network characteristics
 - *Distance*
 - *Clustering*
 - *Centrality*
 - *Connected components*
 - *Community*
 - *Link Prediction*

Network

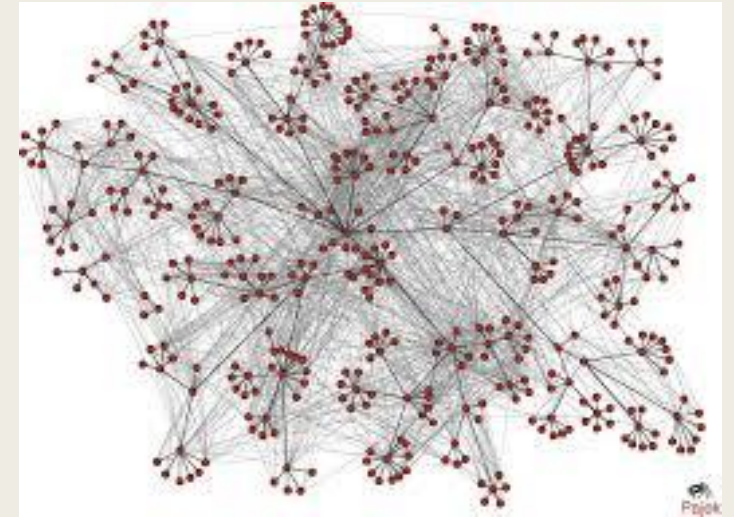
A set of objects (nodes) with
interconnections (edges)



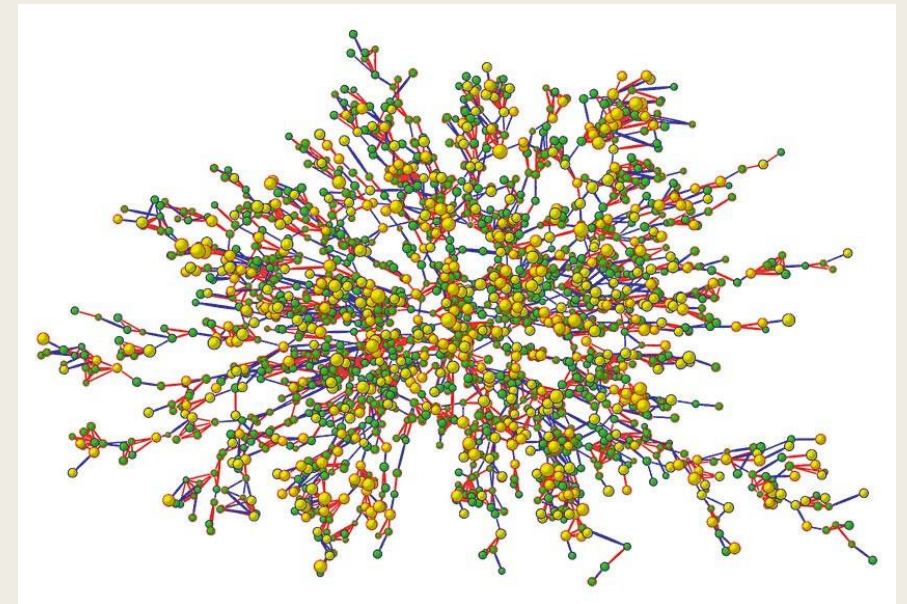
Social Networks



Zachary's Karate Club ([Wikipedia](#))

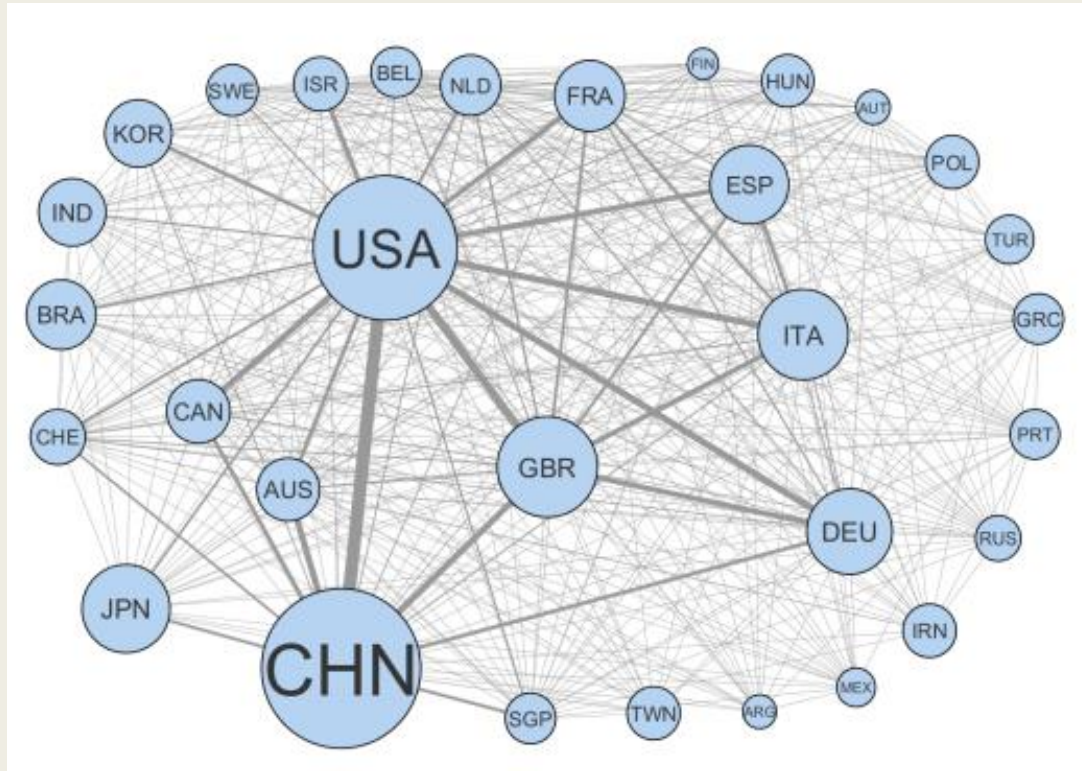


Email Communication Network ([Adamic and Adar, 2005](#))



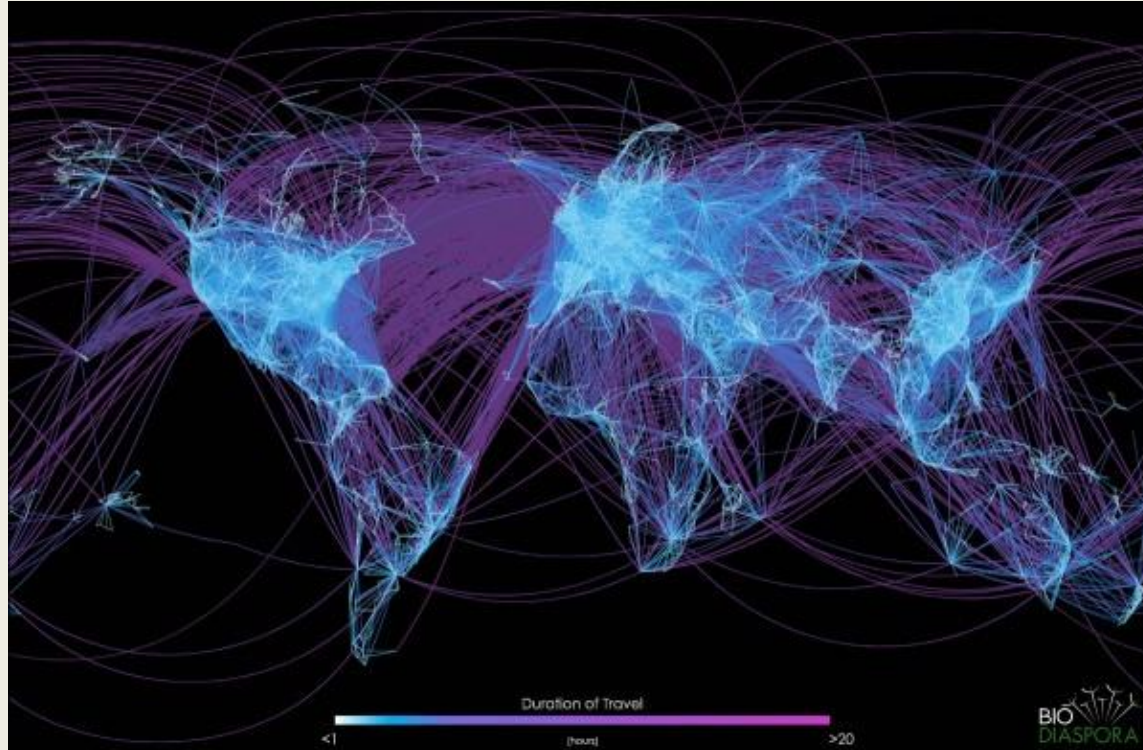
Spread of obesity in social network ([Christakis and Fowler 2007](#))

Co-authorship Network



Network science collaborations by country of authors ([Molontay and Nagy, 2019](#))

Transportation Network



Air Travel ([Visualizing Data](#))



Other Networks

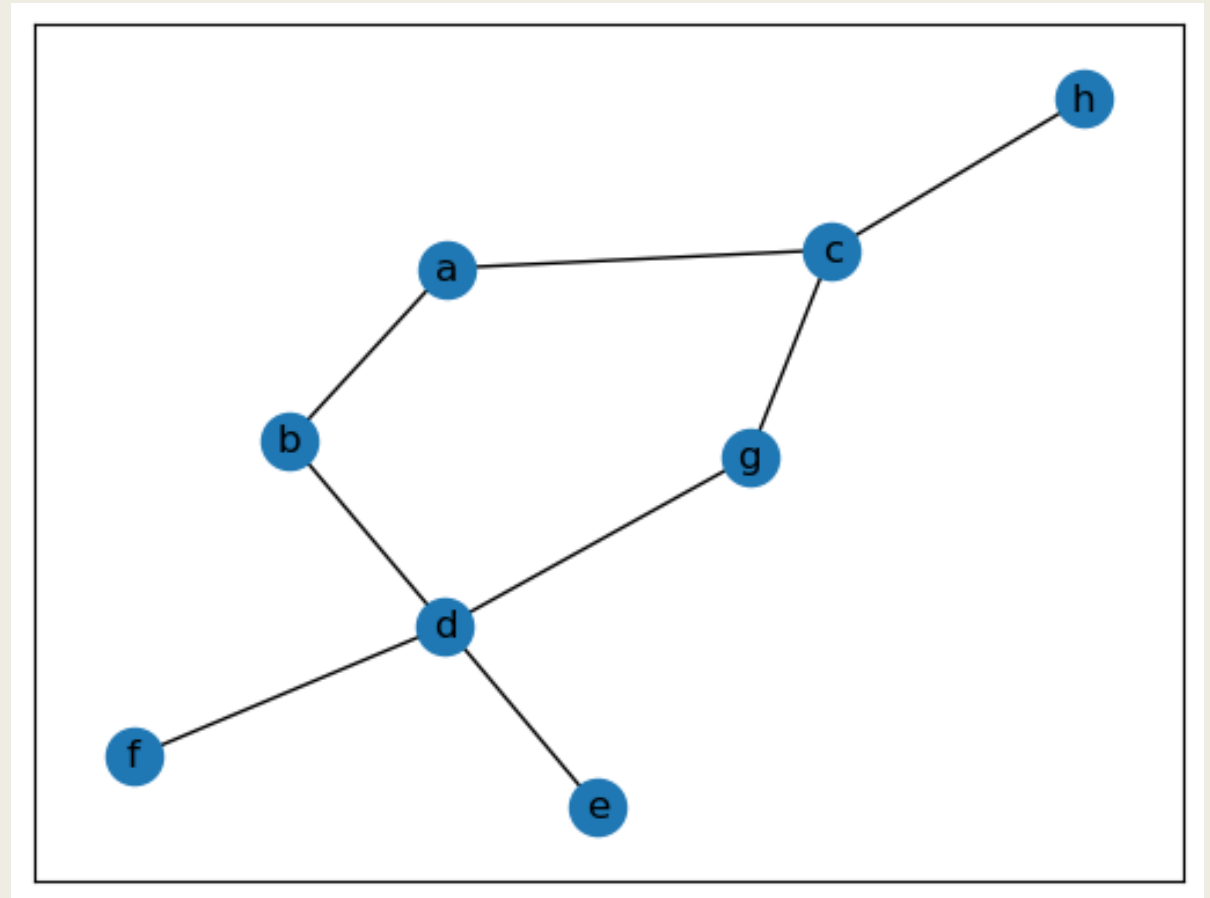
- Biological: Interactions between genes or proteins
- Technological: Internet routers or computers
- Trade networks
- Citation networks
- Financial networks
- ...

Applications

- Social Sciences: Influence, communities, diffusion.
- Biology: Gene/protein interaction networks.
- Computer Science: Web search (PageRank), security.
- Marketing: Viral campaigns, recommendation systems.
- Finance: Interbank lending, systemic risk.
- Epidemiology: Disease tracking with contact graphs.

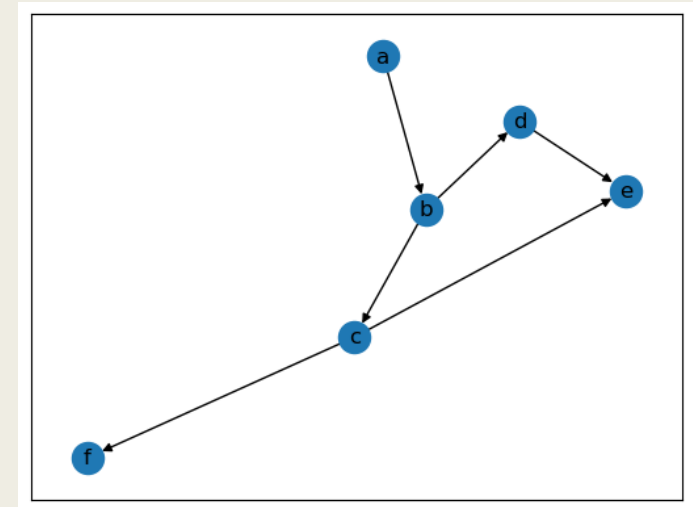
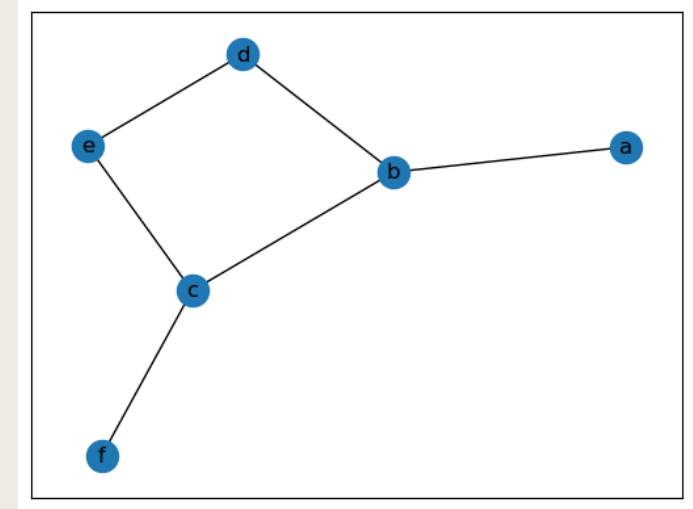
Network Terminology

- Node (Vertex): Entity in the network.
- Edge (Link): Connection between nodes.



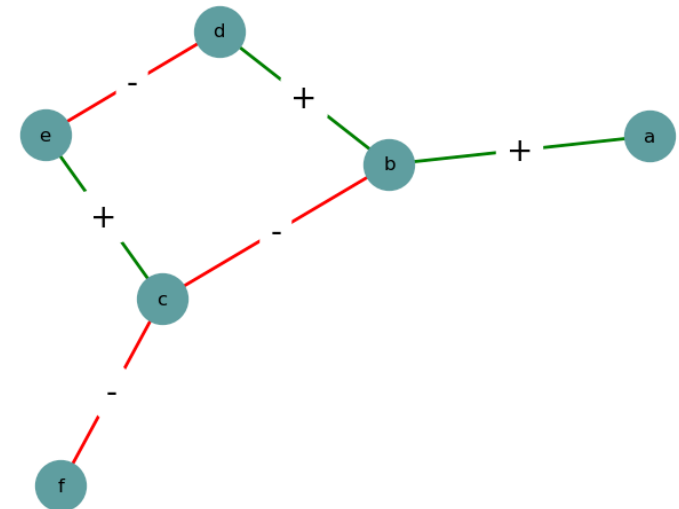
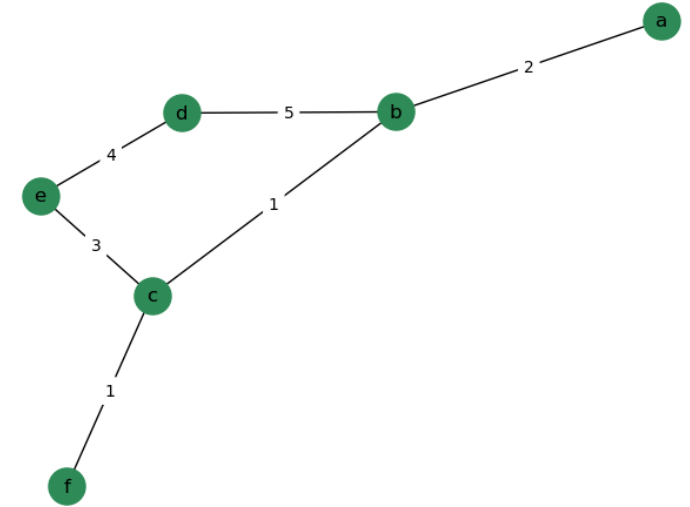
Types of Networks

- Undirected: Edges have no direction.
- Directed: Edges have direction (e.g., Twitter).



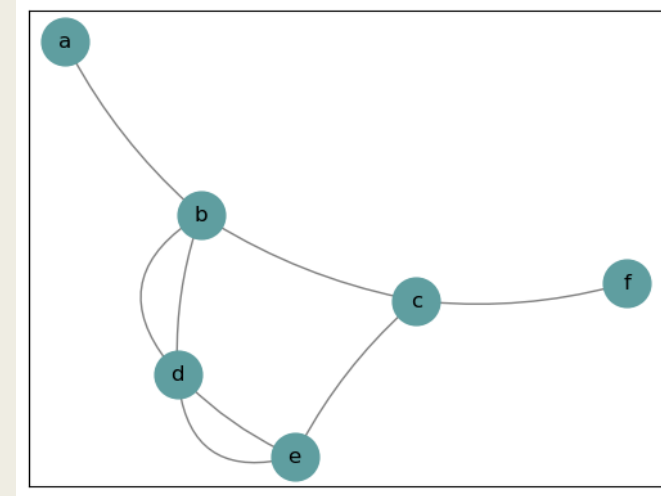
Types of Networks

- Weighted: Edges carry numeric weights.
- Signed: Edges indicate positive or negative relations.

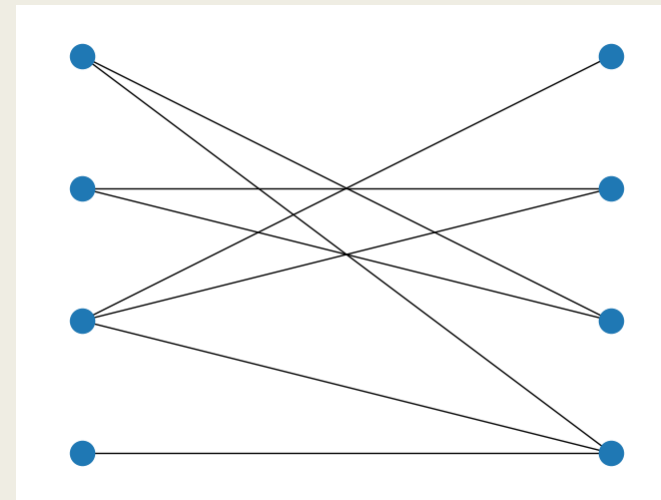


Types of Networks

- Multigraphs: Multiple edges between node pairs.
- Self-loops: Edges from a node to itself.



- Bipartite: Two distinct node sets (e.g., authors and papers).

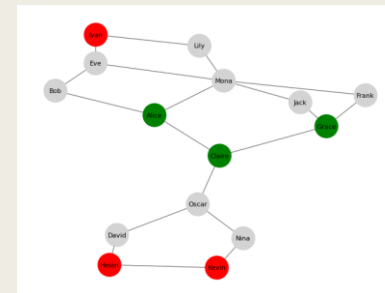


Network Characteristics

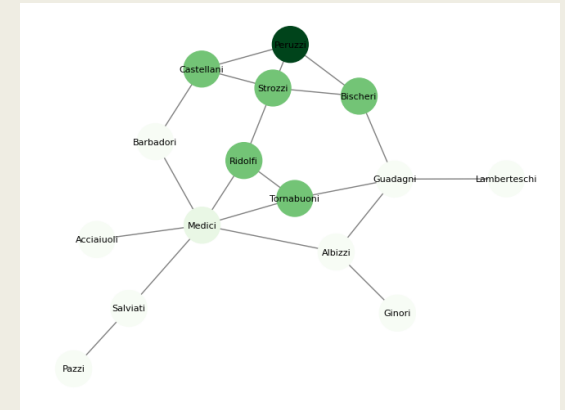
- Distance
- Clustering
- Centrality
- Connectivity
- Community
- Link Prediction

Distance

- Distance reflects how far apart nodes are. Distance metrics help understand how information or influence may travel through the network and how accessible nodes are.
- Pairwise distances
 - *Shortest Path* is the distance between two nodes is the length of the shortest path between them
 - *Eccentricity* of a node n is the largest distance between n and all other nodes
- Summary of Network distances
 - *Average Distance*: Average of all pairwise distances between nodes
 - *Diameter*: Maximum distance between any pair of nodes
 - *Radius*: The minimum eccentricity in the graph
- Central and Peripheral Nodes
 - *Periphery* is the set of nodes with eccentricity = diameter
 - *Center* is the set of nodes with eccentricity = radius



Triadic Closure and Clustering



- Triadic closure is the tendency for people who share connections in a social network to become connected. In other words, the new edges that appear in a network are likely to be the ones that close triangles
 - *“a friend of a friend is likely to become a friend”*
- Clustering Coefficient is a formal way of measuring triadic closure.
 - *Local Clustering Coefficient: Defined for a single node*
 - High local clustering coefficient indicates the node is embedded in a tightly knit group
 - *Global Clustering Coefficient: Aggregate measure over the entire graph*
 - Average clustering coefficient: Unweighted average of all local clustering coefficients
 - Transitivity: Degree weighted average of all local clustering coefficients
 - *A network with high global clustering tends to form communities and exhibits social cohesion*

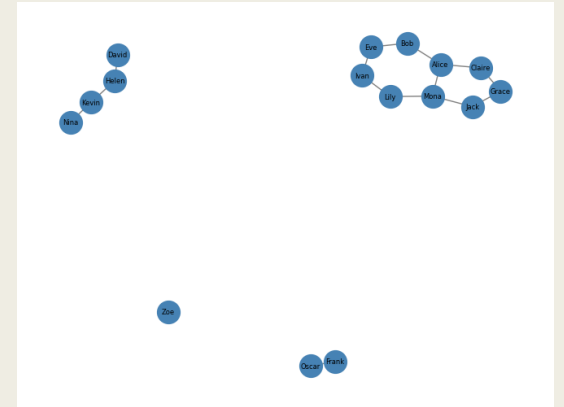
Centrality

- Centrality in network analysis refers to quantitative measures used to identify the most important, influential, or strategically positioned nodes within a network.
 - *in social networks, central individuals can influence opinion, spread information, or broker resources*
 - *in transportation networks, central nodes may serve as hubs or bottlenecks*
 - *in technological networks, centrality can identify vulnerable computers or routers*

Centrality

- Measures of Centrality differ on the assumptions they make about what it means to be "central".
 - *Degree Centrality: Number of connections a node has*
 - *Closeness Centrality: The inverse of the average shortest path from a node to all other nodes. Measures how quickly information can spread from that node.*
 - *Betweenness Centrality: Measures how often a node appears on the shortest paths between other pairs of nodes. These nodes serve as bridges.*
 - *Page Rank: Considers quality of connections*

Connected Components



- In an undirected graph, a connected component is a group of nodes where:
 - *Every node is reachable from every other node in the group,*
 - *And no connections exist to nodes outside the group.*
- In a directed graph, we distinguish between:
 - *Strongly connected components, where every node can reach and be reached from every other node,*
 - *Weakly connected components, where reachability ignores direction.*
- An understanding of connected components in a network can help to
 - *Detect isolated groups in social networks or telecommunication networks.*
 - *Analyze network fragmentation*
 - *Measure reachability*

Communities

- Groups of nodes that are more densely connected to each other than to the rest of the network.
 - *Nodes within the same community are highly interconnected.*
 - *Nodes in different communities have fewer or weaker connections between them.*
- In social networks, communities may reflect friendships, professional groups, or common interests

Link Prediction

- Link prediction aims to estimate the likelihood that a link will form between two nodes that are not currently connected, based on the existing network structure.
Link prediction metrics:
 - *Common Neighbors counts number of mutual friends two nodes share*
 - *Jaccard coefficient normalizes common neighbors by total number of neighbors*
 - *Preferential attachment assumes that those with high degree centrality are more likely to connect*
- Useful in predicting friends in social networks, collaborators in professional networks, or co-authorships in academic networks.

Conclusion

- In this module, we examined
 - *Networks around us*
 - *Applications*
 - *Types of networks, and*
 - *Network characteristics: Distance, Clustering, Centrality, Connected components, Community and Link Prediction*